MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPES 8161 AND 8251

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triodes, ceramic, metal
See figures 1 and 2
Mounting position: Vertical, base down or up
Weight: 8161: 6.25 pounds nominal
8251: 7.5 pounds nominal

ABSOLUTE RATINGS: C telegraphy

TYPE 8161

Parameter:	Fl	Ef	Eb	Ec	Ib	Pg	Pp	Pi	Anode core & seal T	Cooling
Unit:	MHz	Vac	kVdc	kVdc	Adc	W	kW	kW	°C	
Maximum:	110	7.9	4.0	-1.0	2.0	150	4.0	7.2	250	---
Minimum:	---	7.1	---	----	---	---	---	---	---	---
TEST CONDITIONS:	---	7.5	3.0	Adj.	0.83	---	---	---	---	(See note 1)

ABSOLUTE RATINGS: C telegraphy

TYPE 8251

Parameter:	Fl	Ef	Eb	Ec	Ib	Pg	Pp	Pi	Anode Core & seal T	Cooling
Unit:	MHz	Vac	kVdc	kVdc	Adc	W	kW	kW	°C	
Maximum:	30	7.9	4.0	-1.0	2.0	150	4.0	7.2	250	---
Minimum:	---	7.1	---	----	---	---	---	---	-----	---
TEST CONDITIONS:	See note 2									

GENERAL:

Qualification - Required

Ⓑ Denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Filament current		0.65	II	If	49	54	Aac
1231	Peak emission	eb = ec = 2,500 v	0.65	II	is	35	---	a
1261	Electrode voltages (grid)		0.65	II	Ec	-67	-100	Vdc
1266	Total grid current	See note 5	0.65	II	Ic	---	-50	µAdc
1266	Primary grid emission	Ef = 8.3 Vac; t = 15 sec; Ic = 0.9 Adc; anode grounded	0.65	II	Isg	---	-250	µAdc
	Quality conformance inspection, part 2							
1316A	Amplification factor		---	---	Mu	19	26	---
1331	Direct-interelectrode capacitance		---	---	Cgp Cin Cout	16.8 29.2 0.60	23.2 40.2 1.20	pF pF pF
1236	Power oscillation	Eb = 4,000 Vdc; Ib = 1.8 Adc; F = 110 MHz; Rg = 2,150 ohms	---	---	Po	4.2	---	kW useful
	Quality conformance inspection, part 3							
	Life-test	Group C; power oscillation; t = 500 hours	---	---	---	---	---	---
	Life-test end points:							
1231	Peak emission		---	---	is	28	---	a
1266	Primary grid emission		---	---	Isg	---	-250	µAdc
Ⓑ 1143	Forced cooling	Pp = 4.0 kW See notes 3 and 6	---	---	T anode core	--	250	°C
					T fil seal	---	250	°C
Ⓑ ----	Pressure drop	Cooling See notes 4 and 6	---	---	Pressure	---	1.4	In.H_2O

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NOTES:

1. In all electrical tests involving application of filament voltage, forced-air cooling is allowable as follows: base-to-anode airflow, 67 cfm maximum; anode-to-base airflow, 84 cfm maximum through the anode cooler, plus 3 cfm maximum directed into the filament stem structure. These figures apply for air at 40°C maximum and at sea level. At higher air temperature or altitude, suitable correction of the airflow is allowable. Cooling air should be supplied before or simultaneously with application of electrode voltages (including the filament) and may be removed simultaneously with them.

2. The 8251 type may be made by attaching the base flying leads to an 8161 which has met all the requirements listed herein. After attachment of the flying leads and any other mechanical processing, including plating, the 8251 shall meet the requirements of the holding period and the total grid current test.

3. For the cooling test, the tube shall be mounted in an infinite baffle system, or equivalent, so that all air passing in a base-to-anode direction shall pass through the anode cooler of the tube. With cooling air of not less than 20°C nor more than 40°C, both base and anode shall be cooled by applying an airflow of 67 cfm maximum at sea level. At altitudes above sea level, the equivalent mass flow may be used. An appropriate value of anode voltage and current shall be used to produce the specified anode dissipation. Temperatures shall be measured by means of thermocouples, located as follows:

   Anode core: The thermocouple shall be embedded in the base of the anode core, at the base of the cooling fins, by means of drilling a small hole, shallow enough so that the tube vacuum shall not be lost, placing the welded thermocouple junction therein, and then peening the edges of the hole as to hold the thermocouple firmly in place.

   Filament seal: The thermocouple shall be attached, using a suitable cement, to either the inner or outer filament conductor and as close as possible to where the two conductors join the stem dielectric material.

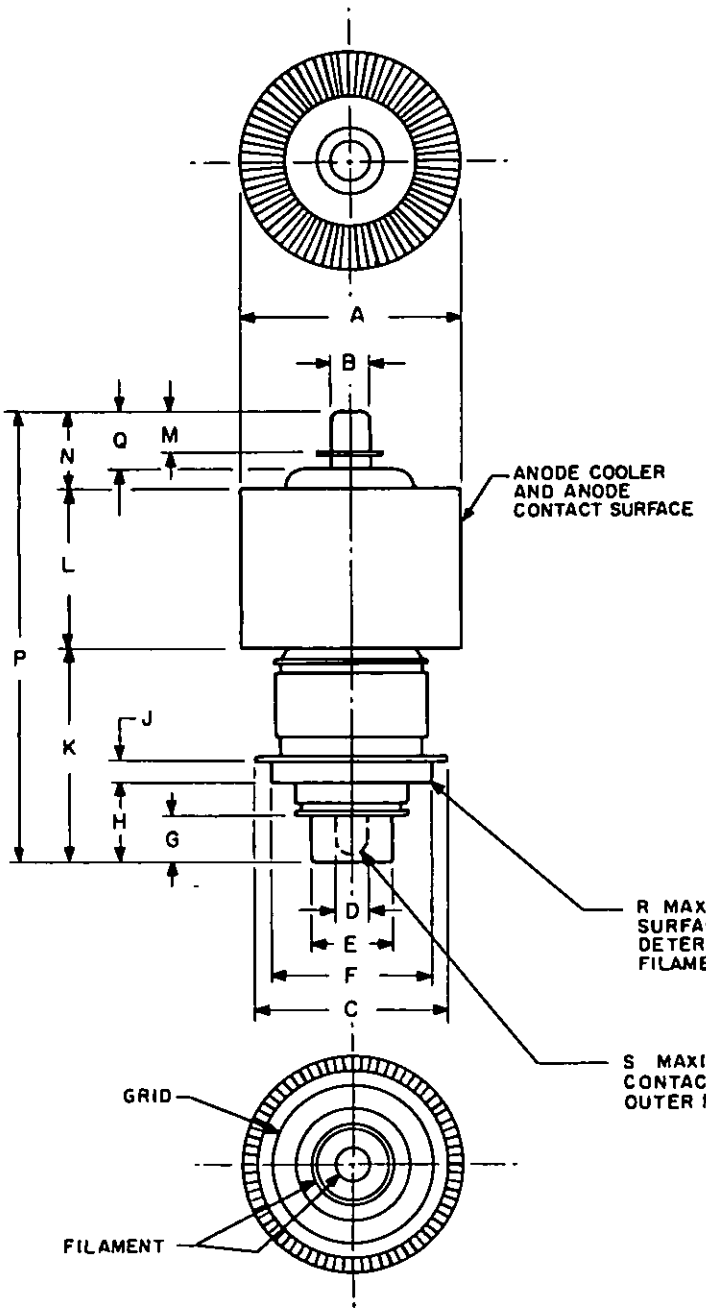
   Good electrical continuity between the thermocouple and the metal surface to which it is attached shall be demonstrated before the cooling test is performed. The size and material of the thermocouples, their installation, and the measuring instrument used to determine temperature shall be in accordance with good engineering practice.

4. The pressure drop across the anode cooler shall be measured while the tube is operating as specified during the cooling test. If test is performed at altitudes significantly above sea level and the airflow is adjusted for an equivalent mass-flow rate, the allowable pressure drop may also be recomputed on the same basis.

5. This test to be performed first at the conclusion of the holding period.

(B) 6. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test shall be made as a part of Quality Conformance Inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar months" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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LTR	DIMENSIONS IN INCHES WITH METRIC EQUIVALENTS (mm) IN PARENTHESES (SEE NOTE a)	
	MINIMUM	MAXIMUM
QUALITY CONFORMANCE INSPECT., PART 2		
A	4.094 (103.99)	4.156 (105.56)
D	.615 (15.62)	.635 (16.13)
E	1.490 (37.85)	1.510 (38.35)
F	2.990 (75.95)	3.010 (76.45)
C		3.625 (92.08)
G	.812 (20.62)	.938 (23.83)
H	1.375 (34.93)	1.625 (41.28)
P	8.000 (203.20)	9.000 (228.60)
R		.040 (1.02)
S		.025 (.64)
QUALITY CONFORMANCE INSPECT., PART 3 SEE NOTE b		
B	.781 (19.84)	.844 (21.44)
J	.375 (9.53)	.438 (11.13)
K	3.875 (98.43)	4.250 (107.95)
L	2.937 (74.60)	3.063 (77.80)
M	.688 (17.48)	.812 (20.62)
N	1.188 (30.18)	1.688 (42.88)
Q	1.000 (25.40)	1.125 (28.58)

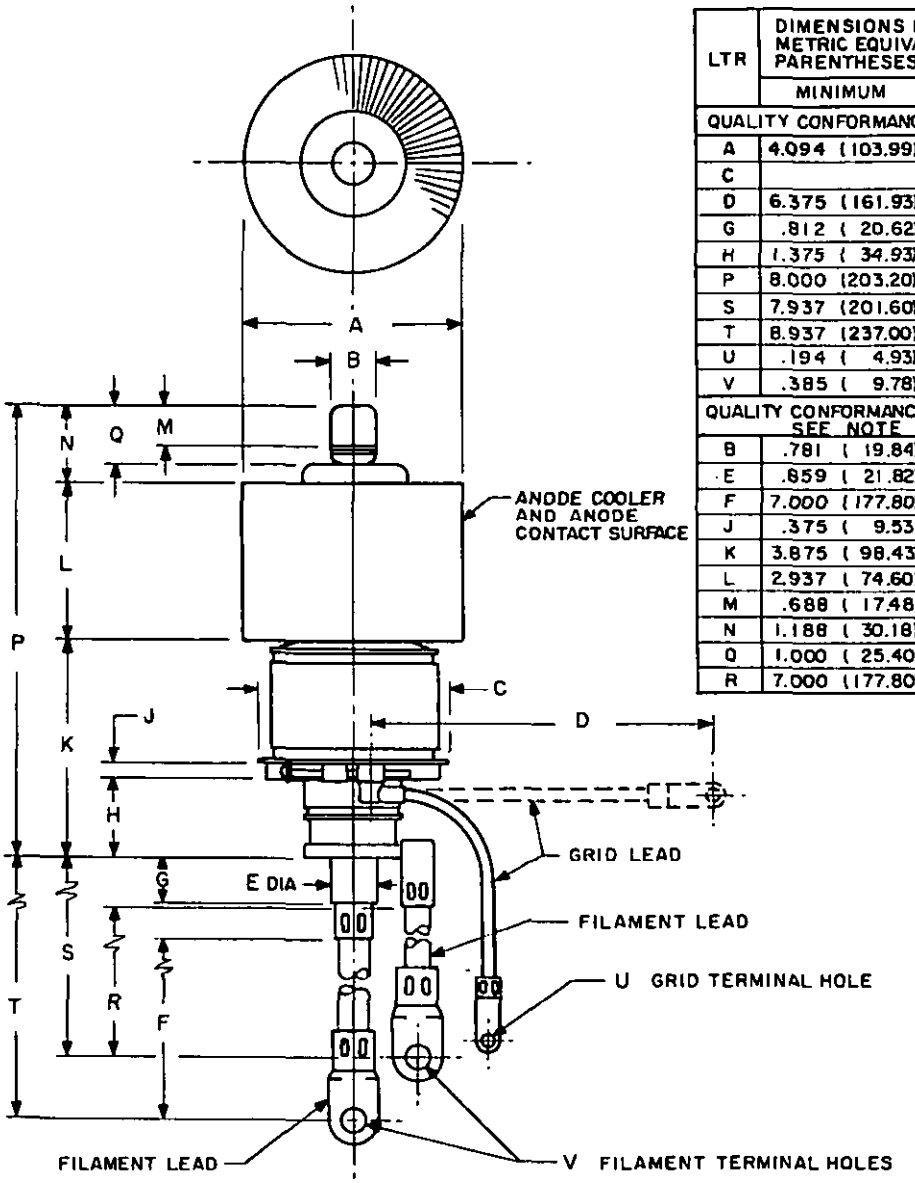
NOTE :-

a. METRIC EQUIVALENTS (TO THE NEAREST .01 mm) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm.

b. SEE NOTE 6.

FIGURE 1. Outline drawing of electron tube type 8161

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LTR	DIMENSIONS IN INCHES WITH METRIC EQUIVALENTS (mm) IN PARENTHESES (SEE NOTE a)	
	MINIMUM	MAXIMUM
QUALITY CONFORMANCE INSPECT., PART 2		
A	4.094 (103.99)	4.156 (105.56)
C		3.625 (92.08)
D	6.375 (161.93)	6.625 (168.28)
G	.812 (20.62)	.937 (23.82)
H	1.375 (34.93)	1.625 (41.28)
P	8.000 (203.20)	9.000 (228.60)
S	7.937 (201.60)	8.437 (214.30)
T	8.937 (237.00)	9.437 (239.70)
U	.194 (4.93)	.200 (5.08)
V	.385 (9.78)	.395 (10.03)
QUALITY CONFORMANCE INSPECT., PART 3 SEE NOTE b		
B	.781 (19.84)	.844 (21.44)
E	.859 (21.82)	.890 (22.61)
F	7.000 (177.80)	7.500 (190.50)
J	.375 (9.53)	.438 (11.13)
K	3.875 (98.43)	4.250 (107.95)
L	2.937 (74.60)	3.063 (77.80)
M	.688 (17.48)	.812 (20.62)
N	1.188 (30.18)	1.688 (42.88)
Q	1.000 (25.40)	1.125 (28.58)
R	7.000 (177.80)	7.500 (190.50)

NOTE:-
a. METRIC EQUIVALENTS (TO THE NEAREST .01 mm) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm.
b. SEE NOTE 6.

FIGURE 2. Outline drawing of electron tube type 8251

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